

Training data

Parameters x + fields

Low fidelity



192 × 384

High fidelity



721 × 1440

Adapted surrogate library

Transfer · M1, M3, M5

Coarse pretraining → fine finetuning

Pooled fidelity · M2

Parameters + fidelity labels

Correction · M4, M6, M8, M9

$x \longrightarrow$ Coarse field $\longrightarrow$ Fine field
Predict / retrieve Learned correction (x)

Reduced basis · M7 (fine only)

$x \longrightarrow$ Coefficients $\longrightarrow$ Fine field
Gaussian process Fixed spatial patterns

Ensemble weight estimation

5 reserved fine pairs (x_i, y_i)

Known y_i and model fields $f_m(x_i)$



Error matrix $\hat{C}$

.5

0

Model weights w_m

Minimize combined fitting error
Nonnegative weights summing to 1
Fixed across inputs and grid points

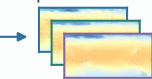
Prediction at a new input

New x

Trained library
M1 to M9

12 emissions
features

9 predicted fields



$$\hat{y}(x) = \sum_m w_m f_m(x)$$

Σ

ERA5 prediction

